throughout our universe, this may have no bearing on the physical attributes in these other universes so long as they are separate from us, or at least so far away that their light has not had time to reach us. And so we can imagine that physics varies from one universe to another. In some, the differences may be subtle: For example, the electron mass or the strength of the strong force might be a thousandth of a percent larger or smaller than in our universe. In others, physics may differ in more pronounced ways: The up-quark might weigh ten times what it weighs in our universe, or the strength of the electromagnetic force might be ten times the value we measure, with all the profound implications that this has on stars and on life as we know it (as indicated in Chapter 1). And in other universes, physics may differ in still more dramatic ways: The list of elementary particles and forces may be completely distinct from ours, or, taking a cue from string theory, even the number of extended dimensions may differ, with some cramped universes having as few as zero or one large spatial dimension, while other expansive universes possess eight, nine, or even ten extended spatial dimensions. If we let our imaginations run free, even the laws themselves can drastically differ from universe to universe. The range of possibilities is endless.

Here's the point. If we scan through this huge maze of universes, the vast majority will not have conditions hospitable to life, or at least to anything remotely akin to life as we know it. For drastic changes in familiar physics, this is clear: If our universe truly looked like the Garden-hose universe, life as we know it would not exist. But even rather conservative changes to physics would interfere with the formation of stars, for example, disrupting their ability to act as cosmic furnaces that synthesize complex life-supporting atoms such as carbon and oxygen that, normally, are spewed throughout the universe by supernova explosions. In light of the sensitive dependence of life on the details of physics, if we now ask, for instance, why the forces and particles of nature have the particular properties we observe, a possible answer emerges: Across the entire multiverse, these features vary widely; their properties can be different and are different in other universes. What's special about the particular combination of particle and force properties we observe is that, clearly, they allow life to form. And life, intelligent life in particular, is a prerequisite even to ask the